



Load Bank Use in Data Centers

White Paper

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Load banks offer practical approaches for testing and maintaining power and cooling systems in data center facilities. This following narrative surveys their use across data center systems and explains their application in IT environments.

Background Information

Data centers play a vital role in supporting modern economic activity, enabling technologies used at industrial and manufacturing facilities as well as in transport, logistics, banking, education, entertainment, and more. Since the onset of the COVID-19 pandemic, digitization has increased due to changes in business-to-business and business-to-consumer commerce, and in work-at-home trends. Consequently, the importance of data centers has never been greater. Their ability to deliver vital services relies on the continuous availability of electrical power.

While the impacts of downtime vary with the type of data center and scale of any outage, downtime costs can commonly amount to hundreds of thousands of dollars per hour. In addition, downtime prevents users from completing vital tasks, which then impacts the reputations of service providers and can incur penalties for not meeting service level agreements. Because power availability is an important part of providing reliable services, verifying power and cooling system capabilities plays an vital role in ensuring service continuity. Load banks provide a means of testing and verifying the performance of data center power and cooling systems to ensure efficient operation and minimize downtime.

Load Bank Use in Data Centers

There are four primary applications where load banks are utilized in data centers:

1. Commissioning
2. Expansion
3. Maintenance and Periodic Testing
4. Equipment Replacement

Load bank requirements for each application vary power, size, type, and technical specification. Ensuring the correct load bank is used is vital to ensuring optimal testing.



Commissioning

In the course of equipment life-cycles, failure rates can be expected to vary according to the relative age of the equipment. Referring to Figure 1, initial failure rates are high early in the life-cycle. The failure rate decreases as defective products are removed and installation errors are corrected, and the equipment operates with relatively low rates of failure thereafter. As products reach the end of their service life, failure rates begin to increase again.

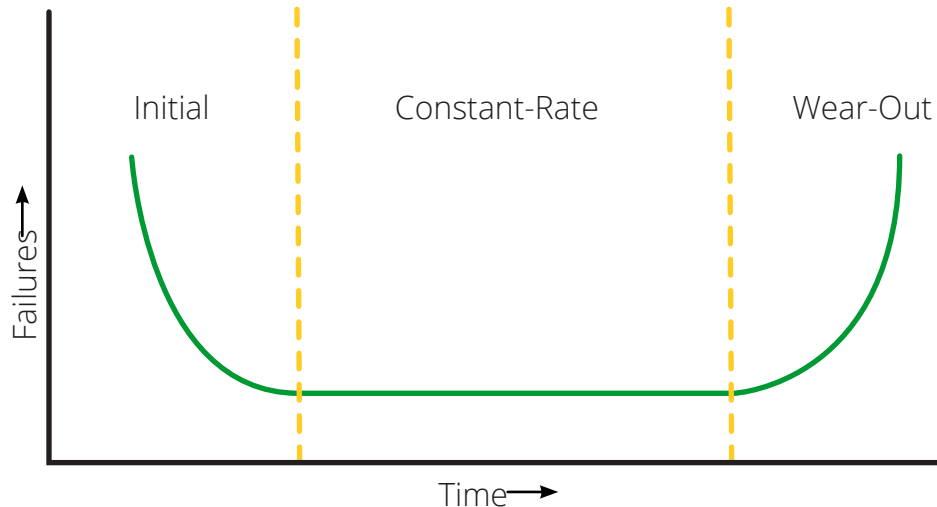


Figure 1: Equipment Life Cycle

Commissioning is the process of ensuring that data system systems and equipment are thoroughly tested when they are installed, and that they operate to the requirements of the end user. Commissioning helps identify potential issues and ensure components and systems work properly and efficiently. During data center commissioning, critical systems are tested, including backup diesel generators; uninterruptible power supplies (UPS); heating, ventilation, and cooling (HVAC) systems; electrical bus and distribution cabling; power distribution units; and remote power panels. Load testing during commissioning ensures that faulty components or systems are identified and repaired or replaced before the facility is turned over to the end-user. Because full facility load is not always available during construction and commissioning, load banks are used to supplement loads to simulate design conditions and verify proper function. For information about testing battery and UPS systems, review [Direct Current Load Banks for Battery Capacity Testing](#).

Load banks are also used to commission data center cooling systems. By placing resistive load banks where heat loads will ultimately be present, cooling systems can be tested to verify their capacity, and to adjust the balance of air flow distribution to optimize cooling performance. Testing cooling performance throughout an entire server room can require locating a quantity of load banks throughout the space. This type of testing is streamlined by networking the load banks to provide common control and data collection functions. Networked load bank controls enable convenient operation and adjustment of load banks during tests. Automated data collection streamlines the evaluation and reporting of load test results. For information about testing data center cooling systems, review the Avtron white paper entitled [Applying Loads in Limited Access Facilities and Distributed Environments - Applications for Suitcase-Style Load Banks](#).

Expansion

Data centers are often sized to accommodate future expansion. When expansion occurs, new systems and components must be tested under load to prove their operability and efficiency. Correct simulation of the newest power and heat loads is necessary. Load banks are used to execute the required testing. For example, a data center may add a number of server racks to an existing facility. Co-location customers may require evidence of the performance of the power and cooling systems serving these locations. Multiple load banks can be used to simulate the electrical load that will be placed on the power distribution systems by the new equipment. Resistive load banks can be placed at key locations to verify local cooling system performance. The resulting data can be automatically compiled into reports that verify compliance to customers purchasing server space.

Maintenance and Periodic Testing

Most data centers utilize both backup UPS systems to support computing during relatively short power interruptions and diesel generators to supply power for longer outages. Regular maintenance and testing of these systems are vital to ensuring their availability. Load banks effectively test and stress electrical systems for maintenance and periodic testing purposes. In these applications, a variety of load test types can be configured for specific maintenance purposes. The resulting reports identify issues that can then be resolved. For additional information, review the Avtron resources referenced in the Commissioning section of this document.

Equipment Replacement

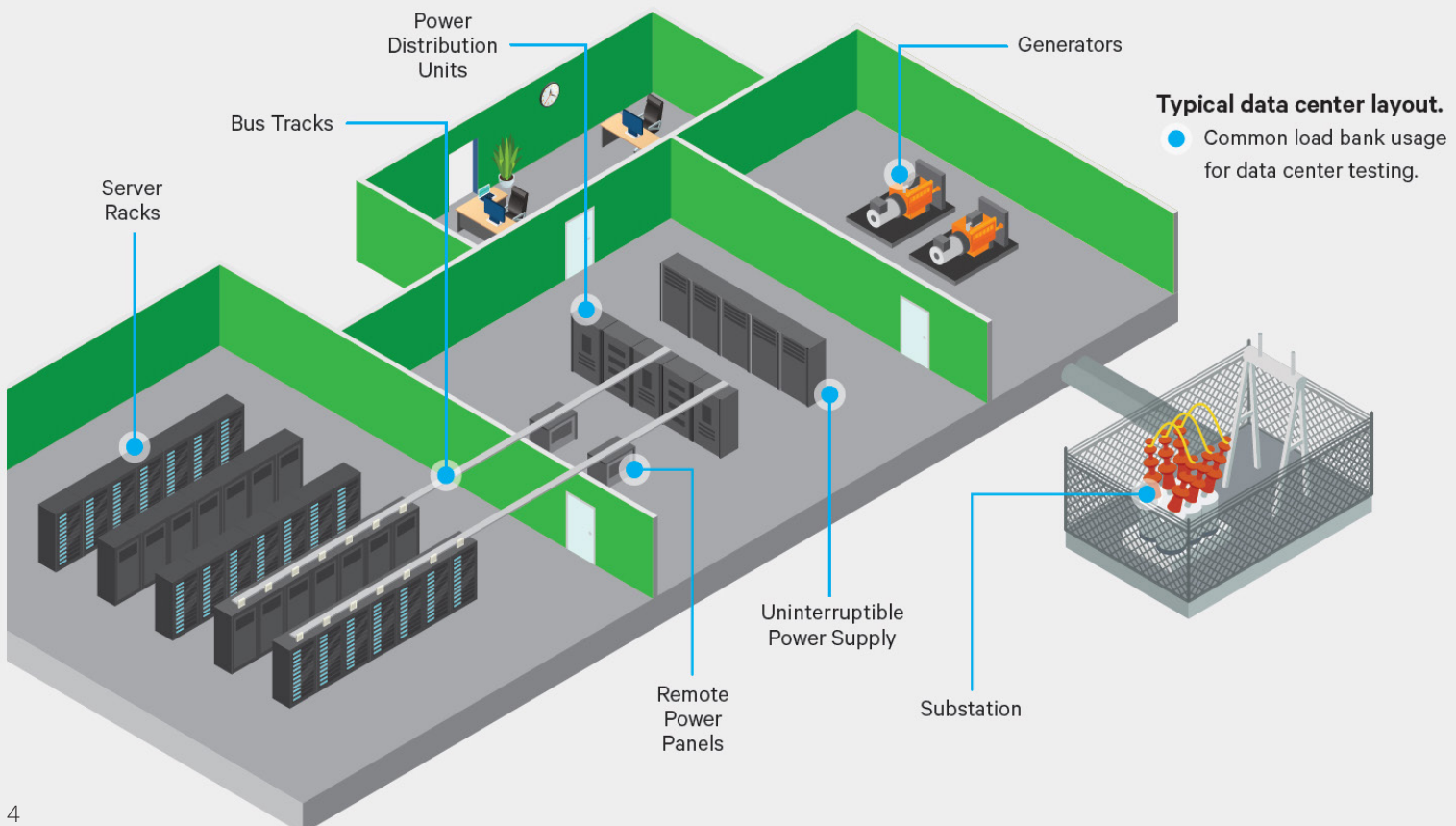
When specific components or equipment are replaced or repaired, their proper function should be verified. Load testing is effective for verifying the proper function and performance of repaired or replaced equipment prior to return to service.

Types of Load Banks for Data Center Systems

Load bank elements apply resistive, inductive, or capacitive loads to circuits, or a combination thereof. When testing data center systems, load banks should be selected to provide the type of load that best matches the characteristics of the load equipment. For instance, circuits serving primarily motor loads should be tested using an inductive load bank. For more information about the types of loads that can be applied, review the Avtron article entitled [Resistive, Inductive, and Capacitive Load Banks](#) and the Avtron white paper entitled [Resistive, Inductive and Capacitive Load Elements](#).

Figure 2 illustrates the configuration of backup power system in a data center facility. For an interactive 3D Model of backup power equipment in a data center, review the Avtron Power Solutions [Interactive 3D Facilities](#).

Figure 2: Backup Power System Configuration



Application	Type
Backup Generators	Resistive and Inductive Load Banks
Uninterruptible Power Systems	Resistive Permanent Load Banks
Heating, Ventilation, and Air Conditioning Systems	Resistive Portable Load Banks
Substation	High Voltage Load Banks
Power Distribution Units	Resistive Portable Load Banks
Remote Power Panels	
Bus Work	

The table lists types of load bank typically used for select applications.

Backup Diesel Generators

The majority of data centers use one or more onsite diesel generators to supply power during utility outages. Correct operation and maintenance of the diesel generator is vital in ensuring maximum uptime of a data center.

Load bank testing verifies generator availability and performance. A load bank will simulate electrical load and allow testers to identify potential problems. This enables correction of conditions that are found so that backup power will be available when required. A load test can verify the correct operation of an engine-generator's voltage regulator, governor, cooling system, fuel system, and control system.

Applying load also avoids diesel "wet stacking" that occurs when gensets are operated with insufficient load. Doing so can cause unburned fuel residues to accumulate in or on engine combustion chambers, injector nozzles, piston rings, turbochargers, and exhaust systems. The result is diminished engine output and potential impacts to reliability. For additional information, review the Avtron article entitled [Manage Load To Avoid Diesel Wet Stacking](#).

Either a purely resistive or a resistive-inductive load bank can be used to test diesel gensets. A purely resistive load bank will evaluate an engine's ability to provide the rated kW and deliver an equivalent amount of kVA. A resistive-inductive load bank will fully test the power supply by applying impedance currents out-of-phase with voltage.

Diesel generators should be load tested periodically in accordance with regulatory and industry codes as well as customer requirements. In larger data centers where loads are greatest, permanently installed, containerized, resistive inductive load banks are often used to provide sufficient load and endure outdoor environmental conditions.

Many load bank suppliers offer control systems that perform automatic and transient load tests. The load test results can be archived and reviewed assess generator performance over time. Alternatively, load bank control can be integrated into existing control systems often found in larger data centers.

HVAC Systems

Insufficient cooling can impact the reliability of servers and other IT equipment. Load banks are an important part of HVAC commissioning activities. Because load banks dissipate electrical power as heat, they are used to simulate the heat output of IT equipment. An HVAC system's reaction to this heat can be monitored to test the effectiveness of the cooling system. In the course of this work, load banks can be moved to specific locations to test local HVAC performance. Networking these load banks enables someone to operate all units from a single handheld control to save operator time and ensure consistent load. An operator can adjust the operation of individual load banks to create hot spots as needed.



UPS Systems

In its simplest form, a UPS is an enclosed assembly of battery packs that automatically supplies power to loads if mains power fails. UPS systems are especially useful for providing power from the onset of a mains outage until acceptable power is supplied by the facilities backup power generation system. A UPS may only need to run for a few seconds or minutes; however, bumpless power source transition can be vital to the operation of downstream IT loads.

Load testing validates the correct operation and performance of a UPS system. One or more load banks are used to discharge the batteries in accordance with manufacturer specifications and evaluate their ability to supply the necessary amount of power. In addition, the electrical performance of UPS invertors and distribution equipment can be evaluated to ensure correct operation.

Typically, UPS systems will be tested as part of a scheduled maintenance program with other back-up power supplies such as diesel generators. Resistive load banks are typically used for UPS testing.

Substation

The power provided to a data center by a substation must be both reliable and adequate to meet IT power demands. Load banks are routinely used to test the capacity of substations to provide rated power and verify the performance of relays and meters. A load bank can also be used with a transformer to test medium and high voltage feeds.

Power Distribution Units

Power Distribution Units (PDUs) may be rack-mounted and are used control and monitor power to servers, switches, and other devices. Resistive load banks are often used during commissioning to test PDUs and verify correct operation. Because a PDU can often be sized at 100 to 300kVA, portable castor-mounted load banks are typically used for testing them. Portable load banks can be positioned at each PDU in the data center hall, then power can be applied and PDU operation can be monitored. Load testing PDU's is a cost-effective method of ensuring that multiple racks will receive the power for required by downstream equipment.

Remote Power Panels

Remote Power Panels (RPPs) are used in conjunction with PDUs. The RPP is a separate enclosure that allows a sub-feed from a PDU to connect a distribution panel that feeds load equipment. The RPP enables PDUs to be placed in a separate room, and also allows larger PDUs with sub-feed breakers to be used. Load banks are regularly used to commission RPPs to ensure proper operation and distribution of power from the PDU to the rack systems.

Bus Tracks

Bus track power distribution is becoming more common in data centers, allowing greater flexibility and accommodation of rising power densities. Load banks test bus track power units, feed tracks, and tap offs to ensure efficient operation.



Summary

Data Centers provide vital services that rely on the continuous availability of electrical power. Because power availability is essential to providing reliable services, verifying power and cooling system capabilities plays an important role in ensuring service continuity. Consequently, load banks are used to commission, expand, maintain, and test power and HVAC systems and equipment, including replacement equipment. These systems include backup generators, HVAC and UPS systems, substations, power distribution units, remote power panels, and electrical bus work.

In order to test electrical systems and equipment, load banks must be selected to apply the appropriate types of loads to power sources. A range of options is summarized in the table herein. Testing can be streamlined by using load bank control systems to record data and compile test reports. Networking multiple load banks together enables control of many units from a single electronic controller for further productivity gains.



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